together individual threads, the raw material of common textiles. Similarly, we can ask ourselves whether there is a raw precursor to the fabric of spacetime—a configuration of the strings of the cosmic fabric in which they have not yet coalesced into the organized form that we recognize as spacetime. Notice that it is somewhat inaccurate to picture this state as a jumbled mass of individual vibrating strings that have yet to stitch themselves together into an ordered whole because, in our usual way of thinking, this presupposes a notion of both space and time—the space in which a string vibrates and the progression of time that allows us to follow its changes in shape from one moment to the next. But in the raw state, before the strings that make up the cosmic fabric engage in the orderly, coherent vibrational dance we are discussing, there is no realization of space or time. Even our language is too coarse to handle these ideas, for, in fact, there is even no notion of before. In a sense, it's as if individual strings are "shards" of space and time, and only when they appropriately undergo sympathetic vibrations do the conventional notions of space and time emerge.

Imagining such a structureless, primal state of existence, one in which there is no notion of space or time as we know it, pushes most people's powers of comprehension to their limit (it certainly pushes mine). Like the Stephen Wright one-liner about the photographer who is obsessed with getting a close-up shot of the horizon, we run up against a clash of paradigms when we try to envision a universe that is, but that somehow does not invoke the concepts of space or time. Nevertheless, it is likely that we will need to come to terms with such ideas and understand their implementation before we can fully assess string theory. The reason is that our present formulation of string theory presupposes the existence of space and time within which strings (and the other ingredients found in M-theory) move about and vibrate. This allows us to deduce the physical properties of string theory in a universe with one time dimension, a certain number of extended space dimensions (usually taken to be three), and additional dimensions that are curled up into one of the shapes allowed by the equations of the theory. But this is somewhat like assessing an artist's creative talent by requiring that she work from a paint-by-number kit. She will, undoubtedly, add a personal flair here or there, but by so tightly constraining the format of her work, we are blinding ourselves